

6(a)	xylem vessel elements ; A singular A vessel elements A xylem elements	1
6(b)	any two from: transpiration pull ; A water, drawn up / pulled up owing to / AW, water evaporating from spongy mesophyll surfaces / water vapour lost from, leaf / plant, during transpiration ; <i>ref. to transpiration not required if transpiration pull stated</i> (overall) water potential gradient ; <i>in context of creates pulling force</i> <i>low water pressure at top and high water p at bottom</i> A pressure gradient A hydrostatic pressure gradient cohesion <u>and</u> adhesion (of water molecules) / cohesive and adhesive forces (of water) ; A cohesion between water molecules or adhesion between water molecules and (xylem vessel) walls / cellulose / hydrophilic parts of lignin	2

Question	Answer	Marks
6(c)(i)	any two from: 1 (lignified wall / lignin) prevents, (inward) collapse / compression / AW (from tensions created) ; R bursting 2 (lignified wall / lignin) rigid / strong / secondary thickening, for support (of xylem vessels) ; R support, stem / plant 3 adhesion of water to hydrophilic groups in lignin, to maintain column of water / for water pathway / for transpiration stream / AW ; 4 waterproofing properties (of lignin) / AW / water prevented from leaving xylem vessels (supports idea of continuous columns) ; 5 lignification, associated with cell death / allows formation of hollow cells / allows formation of hollow tubes (for minimum resistance to flow) ; 6 AVP ; cell walls rigid to allow tension to develop	2
6(c)(ii)	any one from: cell death / apoptosis ; loss of cell contents / become hollow ; A loss of (all) organelles R idea that some organelles remain loss of cytoplasm ; A loss of protoplasm loss of nucleus ; AVP ; e.g. formation of pits	1